

Adaptive Candidate Selection for Period Recovery Under NISQ Noise

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Abstract

1 Introduction

RSA encryption was first proposed in 1978 as a way to ensure that electronic mail is both private and can be signed [4]. RSA encryption is based off of the notion that multiplying two extremely large prime numbers is easy, but factoring an extremely large number into two prime numbers is a difficult process, or as described by the creators, "a trap-door one-way function" [4]. Since this time, RSA has become one of the main ways of encryption, used in many applications where the security of digital data is required [1]. However, with the advancement of quantum computing, there is a real possibility that sometime in the near future that anything encrypted with RSA will become vulnerable. This so called "Quantum Day", the hypothetical day when quantum computers will be able to break RSA encryption, is hard to predict, as the tools required for this task are still being created and developed [2].

Shor's Algorithm is the main focal point of breaking RSA encryption. It is an algorithm that uses quantum methods and order finding in order to bypass the difficulty that factoring presents [5]. Despite advancements in Shor's Algorithm that require fewer qubits [6], the scale of quantum computer that is required to break RSA encryption is currently unavailable. The other problem with the algorithm is that it is highly susceptible to Noisy Intermediate-Scale Quantum (NISQ) technology, which all quantum computers currently are. Noise is currently a major problem for quantum computers that possess 50 qubits or more [3]. As such, methods to help mitigate the noise problem are currently being investigated. One of these methods, an approach that classifies features of the noisy distribution that is provided by Shor's Algorithm in order to figure out if the process is recoverable despite the noise [7]. We will examine this methodology

and expand upon the novel approach.

2 Background

The main concept of Shor's Algorithm is the quantum order finding methodology. Given an integer N and a random integer that is coprime to N , defined as a , Shor's algorithm attempts to find the smallest positive integer r , where:

$$a^r \equiv 1(\text{mod}N) \tag{1}$$

Once r , also known as the hidden period, is obtained, it is relatively simple to extract the real factors using Euclid's greatest common divisor algorithm.

$$\begin{aligned} p &= GCD(a^{r/2} - 1, N) \\ q &= GCD(a^{r/2} + 1, N) \end{aligned} \tag{2}$$

This order information is encoded in the measurement distribution of a t -qubit phase-estimation, also known as precision, register. With the ability to write $Q = 2^t$ unique bitstrings which can hold any value of y that is between 0 and $Q - 1$ for the measured register value, the output is a probability distribution $p(y)$. In an ideal, noiseless environment, $p(y)$ should be concentrated and evenly spaced, with the concentrations near the locations of $y \approx Q * s/r$, where $s = 0, \dots, r - 1$ [7].

The problem is that when noise is taken into account, the ideal structure becomes distorted, thus muddling the hidden period. Sometimes, this leads to situations where the correct hidden period is nonrecoverable.

3 References

References

- [1] Boneh, D. Twenty Years of Attacks on the RSA Cryptosystem. Notices of the AMS, 46(2), 203-213. 1999.
- [2] Mosca, M. Cybersecurity in an Era with Quantum Computers: Will We Be Ready? IEEE Security & Privacy. 2018.
- [3] Preskill, J. Quantum Computing in the NISQ Era and Beyond. Quantum 2. 79. 2018

- [4] Rivest, R.L., Shamir, A., Adleman, L. A Method for Obtaining Digital Signatures and Public-Key Cryptosystems. *Communications of the ACM*, 21(2), 120-126. 1978.
- [5] Shor, P. W. Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer. *SIAM Journal on Computing*. Oct, 1484–1509. 1997.
- [6] Zalka, C. Shor’s algorithm with fewer (pure) qubits. <https://arxiv.org/abs/quant-ph/0601097>. 2024.
- [7] Yang, Q., Markidis, S. When Noisy Quantum Order Finding Remains Recoverable for Shor’s Algorithm. <https://arxiv.org/abs/2605.16074>. 2026.